

MMI714: Homework — Distribution Distances in 1D

Middle East Technical University

Due: Monday, November 7, 23:59 (ODTUClass)

Introduction. This homework helps you *see* how three distances behave on simple 1D data: **KL**, **Jensen–Shannon (JS)**, and **Wasserstein-1 (W1)**. You will compare basic distributions, visualize histograms and CDFs, and comment on when metrics agree or disagree. Keep code minimal; focus on clear figures and short explanations.

Problem 1. Two simple sources (warm-up). Create two 1D samples (each $N \geq 5,000$): pick any *Gaussian* distribution and any *non-Gaussian* distribution (e.g., a two-peak mixture, uniform, or Student- t). Plot *normalized histograms* on the same axis using the *same bin edges*. Write 1 short paragraph describing visible differences (location, spread, tails, or modes).

Problem 2. KL, JS, W1 under different binnings. Using the *same* bin edges for both distributions from Problem 1, compute for *three bin counts* (e.g., 20 / 50 / 100):

$$\text{KL}(P\|Q), \quad \text{KL}(Q\|P), \quad \text{JS}(P, Q), \quad W_1(P, Q).$$

Use a small *additive smoothing* (state its value) to avoid $\log 0$ in KL/JS. Present a *table* (rows = bin counts; columns = distances). Add 2–3 sentences: how did binning/smoothing affect the numbers?

Problem 3. “Metrics disagree” counterexample. Tweak or design a new pair so that *either* (i) W_1 is large while JS is small *or* (ii) JS is large while W_1 is moderate. Provide an overlay histogram and a short explanation of *why* the disagreement occurs (e.g., small shift vs. support gap vs. mode drop). Include $\text{KL}(P\|Q)$ and $\text{KL}(Q\|P)$ in your table and comment on *asymmetry*.

Problem 4. CDF lens for Wasserstein. For one pair from Problems 1–3, plot the *empirical CDFs* and the curve of $|F_P(x) - F_Q(x)|$ over x . Report W_1 and explain, in a few sentences, how the CDF picture corresponds to the value.

Problem 5. Sample-size sensitivity. Fix one pair. For $N \in \{200, 1,000, 5,000, 20,000\}$, recompute $\text{KL}/\text{JS}/W_1$ (use your best bin setting from Problem 2). Plot *distance vs. N* for all three metrics on one figure. Write a short paragraph: which metric stabilizes fastest? which is most variable, and why do you think so?

Deliverables.

1. A single notebook or script that produces all figures and tables.
2. A 1–2 page PDF containing:
 - Your tables and plots,
 - A brief discussion for Problems 3–5,
 - A *Methods box* listing: random seed, bin counts, smoothing value (and KDE bandwidth if you used KDE).

Notes. Use `matplotlib` for plots; you may use `scipy.stats` if helpful. Keep wording simple and plots readable (titles, legends, axes). If a KL value becomes infinite without smoothing, report it and show how smoothing fixed it.